Installing OpenCV

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Installing Python (Windows)

Install Python Windows

(Skip if you already have Python installed)

1. Download Python from website:

```
https://www.python.org/downloads/windows/
```

- Click Download Python 3.10.x onwards for 64 or 32 bit depending on your computer.
- You can use another version of Python just make sure is a stable release.

2. Run the Installer

- IMPORTANT: Check the box “Add Python to PATH”.
- Click Install Now.
- Wait for installation to finish.

Installing Python (Windows)

Verify Installation

1. Open a Command Prompt
   - Windows >> Command Prompt
2. Type:

```
$ python --version
```

   - You should see something like

```
Python 3.12.x
```

Installing Python (Ubuntu)

- Python usually comes pre-installed.
- Open a terminal and verify with:

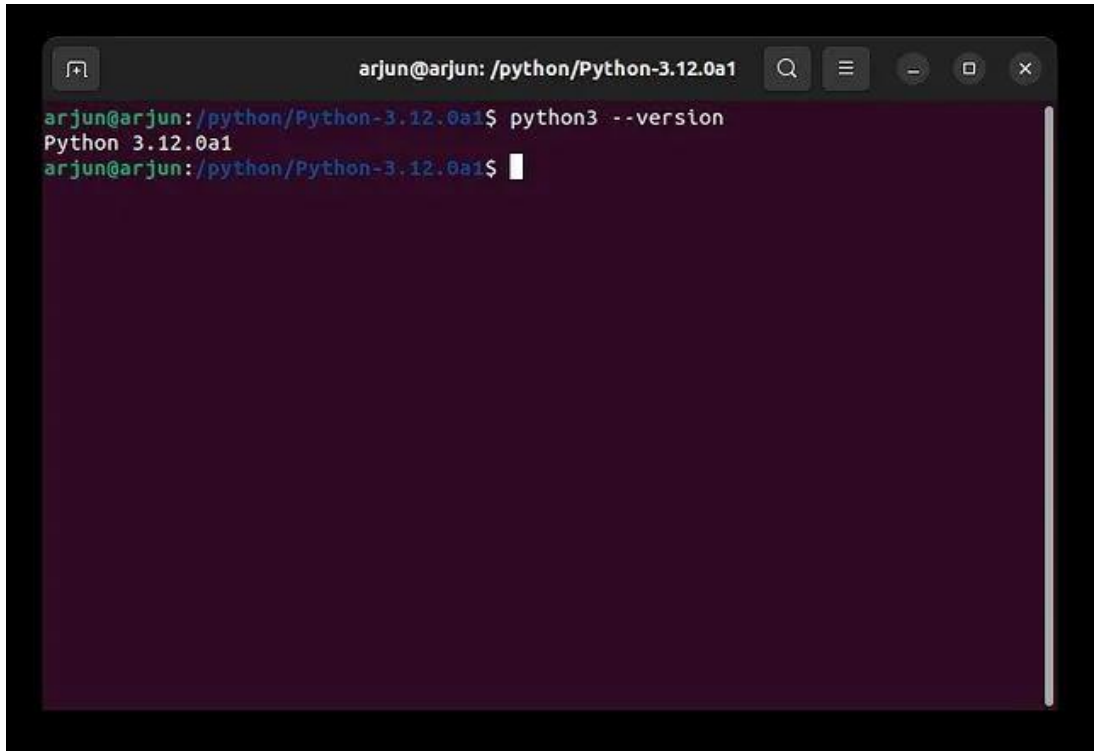
```
$ python3 --version
```

- If missing, you can install it

```
$ sudo apt update
$ sudo apt install python3 python3-pip -y
```

- Verify installation using

```
$ python3 --version
$ pip3 --version
```



Installing Python (Mac)

- Goto:

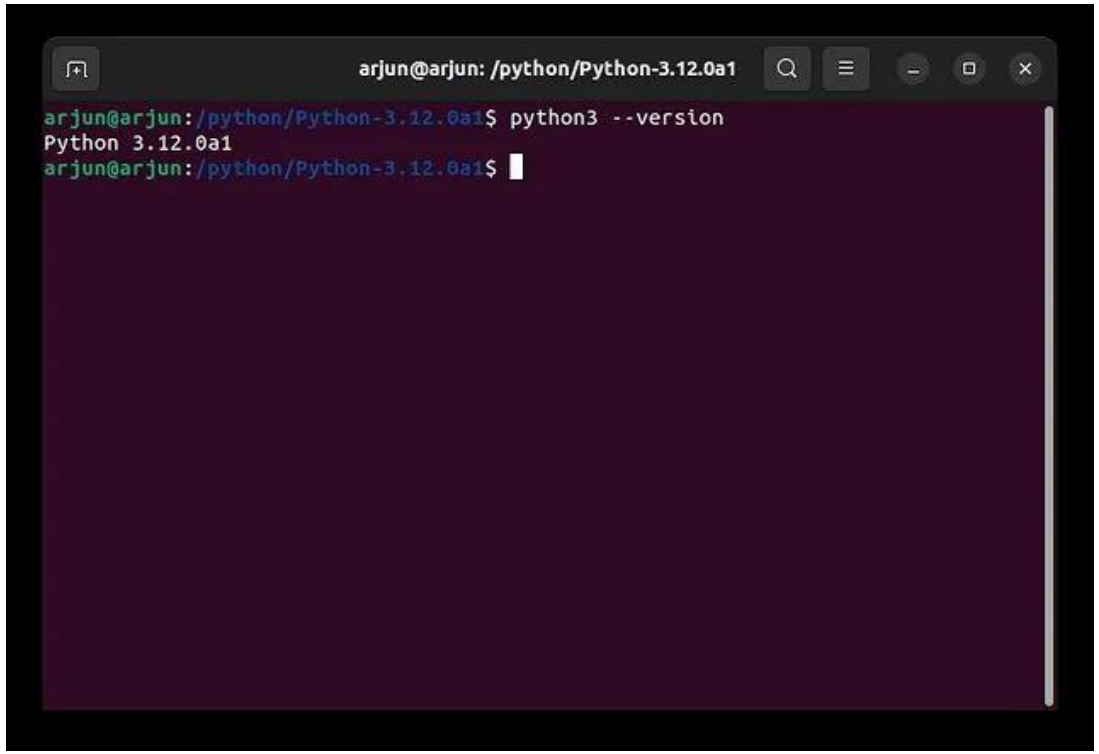
```
https://www.python.org/downloads/macos
```

- Download the latest macOS installer (.pkg)
- Open it and click Continue → Install
- Once done, verify:

```
$ python3 --version
```

- Note: It installs Python under:

```
/Library/Frameworks/Python.framework/
```



OpenCV Installation (Windows)

- Open a cmd prompt and use the following command (make sure you have pip installed):

```
$ pip install opencv-python
```

- Test installation:

```
$ python -c "import cv2; print(cv2.__version__)"
```

- You should see the following

```
PS C:\Users\mario> python -c "import cv2; print(cv2.__version__)"
4.11.0
```

- Please be aware that this is just a basic installation of OpenCV. If you require CUDA or other OpenCV packages such as Aruco, you must install it from source (ubuntu) or using pip in Windows. More information here and here

OpenCV Installation (Ubuntu)

- Open a terminal and use the following command:

```
$ sudo apt update
$ sudo apt install -y python3-opencv libopencv-dev
```

- Test installation

```
$ python3 -c "import cv2; print(cv2.__version__)"
```

- Please be aware that this is just a basic installation of OpenCV. If you require CUDA or other OpenCV packages such as Aruco, you must install it from source (ubuntu) or using pip in Windows. More information here and here

OpenCV Installation (Mac)

- Open a terminal and use the following command:

```
$ pip3 install opencv-python
```

- Test installation

```
$ python3 -c "import cv2; print(cv2.__version__)"
```

- Please be aware that this is just a basic installation of OpenCV. If you require CUDA or other OpenCV packages such as Aruco, you must install it from source (ubuntu) or using pip in Windows. More information here and here

Troubleshoot

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Troubleshooting (Windows)

- Add python PATH (python not found)

[How to Add Python to PATH – Real Python]

- Issue: pip: command not found
  - Windows
    - Your PATH is not configured.
    - Fix: Search: “Add or remove programs" → Python → Modify
    - Select "Add Python to environment variables“
    - Or reinstall Python and check the “Add Python to PATH” box.
  - Ubuntu
    - Install pip

```
$ sudo apt install python3-pip
```

Thank you

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